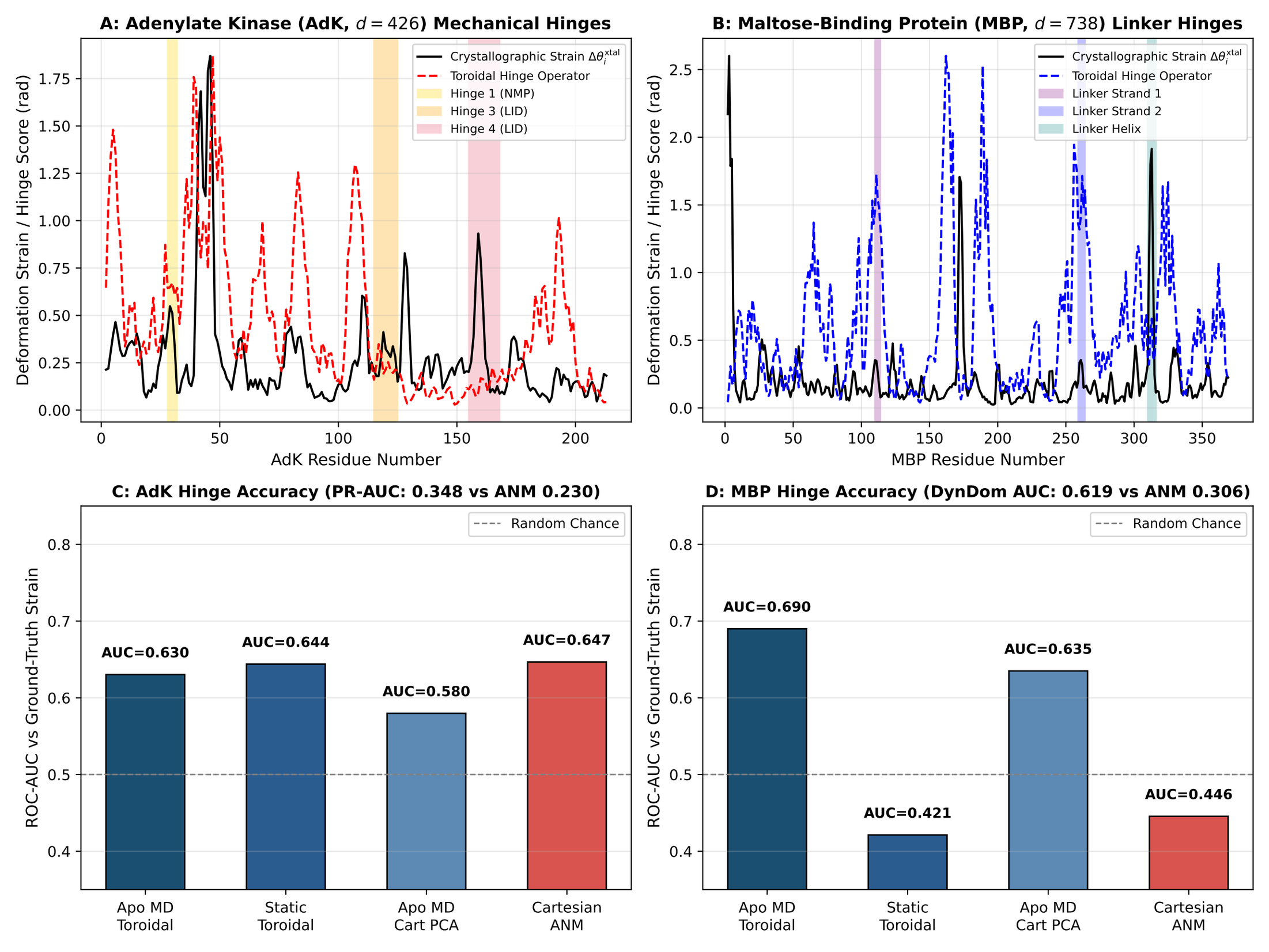
A: Adenylate Kinase (AdK, $d = 426$) Mechanical Hinges
Deformation Strain / Hinge Score (rad)
AdK Residue Number
Crystallographic Strain $\Delta\theta_i^{xtal}$
Toroidal Hinge Operator
Hinge 1 (NMP)
Hinge 3 (LID)
Hinge 4 (LID)
B: Maltose-Binding Protein (MBP, $d = 738$) Linker Hinges
Deformation Strain / Hinge Score (rad)
MBP Residue Number
Crystallographic Strain $\Delta\theta_i^{xtal}$
Toroidal Hinge Operator
Linker Strand 1
Linker Strand 2
Linker Helix
C: AdK Hinge Accuracy (PR-AUC: 0.348 vs ANM 0.230)
ROC-AUC vs Ground-Truth Strain
Random Chance
AUC=0.630
AUC=0.644
AUC=0.580
AUC=0.647
Apo MD Toroidal
Static Toroidal
Apo MD Cart PCA
Cartesian ANM
D: MBP Hinge Accuracy (DynDom AUC: 0.619 vs ANM 0.306)
ROC-AUC vs Ground-Truth Strain
Random Chance
AUC=0.690
AUC=0.421
AUC=0.635
AUC=0.446
Apo MD Toroidal
Static Toroidal
Apo MD Cart PCA
Cartesian ANM